



ECE 501C Project Proposal Banking Database

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Project Proposal Guideline

- Define the system's input-output behavior and the project's scope.
- What is your evaluation metric for success?
- Collect some preliminary data and give concrete examples of inputs and outputs.
- Implement a baseline and an oracle and discuss the gap.
- What are the challenges?
- Which database topics (e.g., normalization, query optimization, transaction management) might be able to address those challenges?
- Search the Internet for similar database systems or projects and mention the related work.
- You should have completed most of the infrastructure setup (e.g., database schema, data collection) and be ready to showcase something interesting.
- In summary, it should include:
 1. problem formulation, clearly identifying the input/output and the challenges
 2. identifying data sources (e.g., public datasets, web scraping, etc.)
 3. literature review with a reference list of relevant database systems or techniques.

Objective & Scope



Objective: Build a bank DB and evaluate based on schema design, query optimization, and SQL anomaly detection



Scope: Database-centric — schema, indexing, latency, materialized views, and benchmarked queries

Bank Schema

Entity	Description
Customer	Stores person info: name, address, contact, authentication credentials (SSN)
Account	Tracks account type (checking, savings, loans), balance, status, account #
Transaction	Record deposits, withdrawals, transfers, timestamps, and involved accounts
Branch	Details about physical branches: location, manager, contact info, routing #
Loan	Loan type, amount, interest rate, term, payment schedule, linked customer
Card	Debit/credit card info, expiration, CVV, linked amount, benefits
Employee	Bank staff details, name, role, assigned branch, salary
Audit Log	Tracks system actions for compliance and fraud detection

Schema Relationships

kaggle

Customers → Accounts → Transactions
 ↓
 Loans

Accounts → Cards

Branches → Employees

Data Schema



 **GitLab**

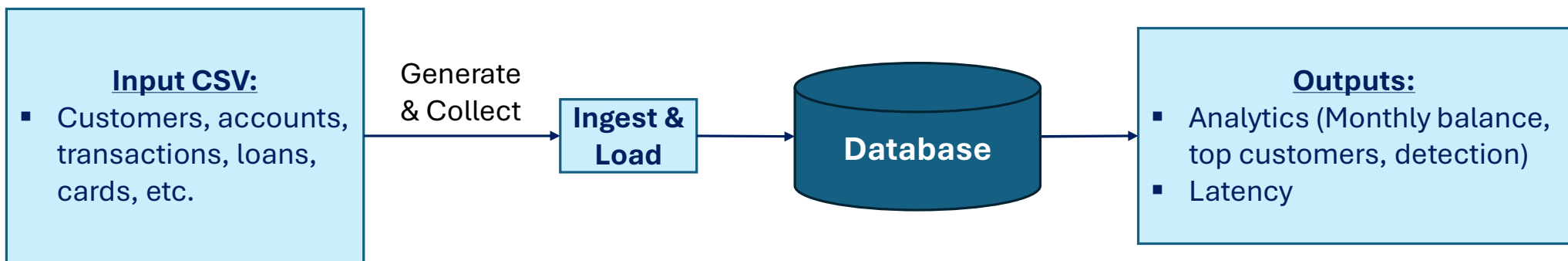
data generator
(numpy + pandas)



Core ER: customers → accounts
→ transactions → cards/loans.



Deliverables: CSVs +
DDL SQL / SQLite.



Baseline vs Oracle



Baseline: raw data & creating DB

1

Optimized: query DB for efficiency (SQL)

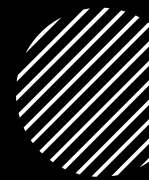
2

Oracle: Best Query with Efficiency & baseline comparisons





Challenges & Mitigations



Scaling large
datasets on a laptop

Mitigation:
stream/generate
data in batches



Long index build time
and storage overhead

Mitigation:
measure index
build time & size



Slow joins and full-
table scans
(transaction-heavy
queries)

Mitigation: TBD
throughout the
semester

Timeline & Milestone Tracker

[illegible]



Thanks! Any Questions Please Email

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